

NIKHIL KAPSE

Unity & XR Developer • Technical Educator • Computer Science Graduate

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SUMMARY

Computer Science graduate and **Unity / XR Developer** with hands-on experience building VR and interactive 3D applications using Unity, C# and Meta Quest 2/3S. I built an independent technical education business around Unity, 3D development and AR/VR, using course development and project-based teaching as a way to **continue building, applying and expanding my technical expertise**. Through this work, I developed 6 practical courses reaching **2,500+ learners through Udemy**, and taught Unity and immersive technologies as Visiting Faculty at MIT institutions in Pune.

Currently, I am working on **MyTvR**, an immersive VR media application involving Unity, C#, Oculus, .NET MAUI, local network streaming and cross-platform client applications. I am now pursuing opportunities in XR/ Unity development, bringing together my **software development background, years of practical Unity/XR development, and technical expertise gained through building real-world projects and technical education** into a professional XR / Unity Developer role.

EXPERIENCE

Independent Unity XR Developer

03/2026 - Present

- Working independently on XR projects, including an immersive Cinema VR application called MyTvR. Currently focused on building production-oriented interactive VR experiences and applying my experience in Unity, C# and XR development to real-world projects, while pursuing full-time opportunities.



AR/VR Development Instructor – Visiting Faculty

MITAOE School of Design

04/2025 - 05/2025 Pune, India

- Invited by MIT institutions in Pune to teach AR/VR, and immersive technologies.
- Instructed class of 20+ design students in AR/VR, enhancing real-world project skills using Unity and immersive technologies.



Unity & XR Technical Educator

Udemy

04/2019 - 02/2026

Udemy Partner Instructor

- Built and operated an independent technical education business focused on Unity, C#, 3D development, game development, creating 6 practical project-based courses that reached 2,500+ learners worldwide through Udemy, with an average rating of 4.3/5 across all courses.
- Achieved approximately 100% year-over-year revenue growth, driven by expansion of course offerings, learner reach, and technical content.
- Built complete technical courses from the ground up, covering curriculum architecture, project development, technical demonstrations, documentation, video production, learner support, and course delivery.
- Supported learners through technical questions and project-related challenges, requiring practical debugging, problem solving, and clear communication of software engineering concepts. Maintained and continuously improved course content, as Unity Workflows & immersive tech evolved.

KEY ACHIEVEMENTS



Udemy Partner Instructor

Built and delivered a portfolio of practical, project-based Unity and AR/VR courses, reaching **2,500+ learners worldwide**.



SIP Abacus Mental Mathematics Competition

1st Runner up in Balewadi Pune (out of 1,000 Participants)
3rd Runner-Up, International Competition (participants from 10 countries), Chennai

EDUCATION

Bachelor of Engineering - Computer Science.

PES's Modern College of Engineering

06/2014 - 05/2018

Junior College : 3rd Rank (Distinction)

PVG's Muktangan Junior College

Date period

Class X : 4th Rank (Distinction)

Vishwakarma Institute Of Technology

Date period

CERTIFICATIONS

Unity VR Development Certification

Issued by : Unity Technologies· [Credential]
Completed Unity's VR Development Pathway, developing practical knowledge of building immersive VR applications, XR interaction systems, locomotion, spatial interactions, and interactive 3D environments using Unity.

Unity Essentials Certification

Issued by : Unity Technologies· [Credential]
Completed Unity's development pathway, strengthening core expertise in Unity's scene architecture, physics, animation, scripting workflows, asset management, and real-time 3D development.

EXPERIENCE



Visiting Faculty – AR/VR

MIT Institute of Design

📅 02/2022 - 03/2022 📍 Pune, India

- Guided Master's-level design students in exploring AR/VR and interactive technologies, providing technical direction and mentorship on immersive design projects. Focused on bridging design concepts with practical implementation using interactive 3D and immersive technology.

PROJECTS

MyTvR / CinemaVR

Unity • C# • VR • Android ExoPlayer • GeckoView • .NET MAUI
Target Platforms: Meta Quest (Android) • Windows PC • macOS

- Developed an immersive VR media platform using Unity and C#, integrating companion .NET MAUI applications for Windows and macOS to enable local-network video streaming between desktop devices and Meta Quest 2.
- Implemented UI object pooling and off-screen element management for media-heavy Scroll Rect interfaces, reducing unnecessary instantiation and active UI rendering overhead while supporting larger media libraries.
- Optimized the Cinema/Theatre environment using lightweight geometry, simplified materials/shaders, static batching, GPU instancing, and baked lighting, maintaining approximately 72 FPS (~14 ms average frame time) during typical VR usage.
- Integrated Android ExoPlayer to overcome Unity VideoPlayer codec and remote HTTP streaming limitations especially for android platform, enabling reliable progressive video delivery from remote MyTvR clients acting as HTTP servers.
- Integrated Mozilla GeckoView to provide embedded browser functionality, enabling access to major web-based streaming platforms directly within the VR application.

RageRoomsVR

Unity • C# • Physics • XR Interaction Toolkit
Target Platforms: Meta Quest (Android)

- Developed a physics-based stress-relief VR experience using Unity, C# and XR Interaction Toolkit, allowing users to pick up and interact with virtual objects using a baseball bat.
- Implemented object pooling for repeatedly respawned environment objects, reusing instances instead of repeatedly creating and destroying GameObjects, thereby reducing runtime GC allocation churn and potential GC pressure.
- Reused particle system instances for repeated impact effects, avoiding unnecessary creation of particle objects and further reducing runtime allocation overhead.
- Implemented physics-based object destruction, impact effects and automatic object respawning to create responsive interactions.
- Managed project source code and version control using GitHub.

RecoVR

Unity • C# • Addressables • VR • Real-Time Rendering
Target Platforms: Meta Quest (Android)

- VR exposure experience using controlled virtual environments designed to help people gradually become more comfortable with fear of heights.
- Implemented Unity Addressables to dynamically load and unload environment sections based on the player's location, controlling active scene content and runtime memory usage.
- Optimized a large city environment using low-poly geometry for distant buildings, detailed models for nearby structures, static batching, and GPU instancing for repeated geometry, and baked lighting to reduce runtime rendering workload.
- Designed the environment with a distance-based level of detail approach.

TECHNICAL SKILLS

Programming C#

OOP

Data Structures

3D Mathematics

Unity 2D & 3D

XR / VR

XR Interaction Toolkit (XRI)

GitHub

Unity Profiler

Performance Optimization

Physics-Based Interaction

Meta Quest

OpenXR

XR Interaction Toolkit

Network Streaming

Cross-Platform Development

VOLUNTEERING

Program Volunteer

Isha Foundation

Volunteered at Isha Yoga Center Coimbatore for Advanced Meditation programs, Project Save Soil, and skill-based volunteer support.

INTERESTS



Trekking/ Hiking



Yoga & Meditation



Organization & Management

ADDITIONAL PROJECTS

PopShotVR • FreakyFrog • Archery Rush • The Church Scene

Please explore all projects by clicking here : [Nikhil's Portfolio](#)